Homework 08 Solutions

Prerith Arunkumar (prerith2)

Margin geometry for a fixed linear score in two and four dimensions, and a peer review of a Homework 7 LDA/QDA analysis skill.

Use of AI tools. I used Claude (Claude Code) to draft the `R` code and exposition for Questions 1 and 2 and to run them. Every number below comes from the code output shown in this report, and I checked the numbers quoted in the text against it. For Question 3, I ran the assigned skill with an AI agent as the assignment requires; the peer review itself is written by me without AI assistance.

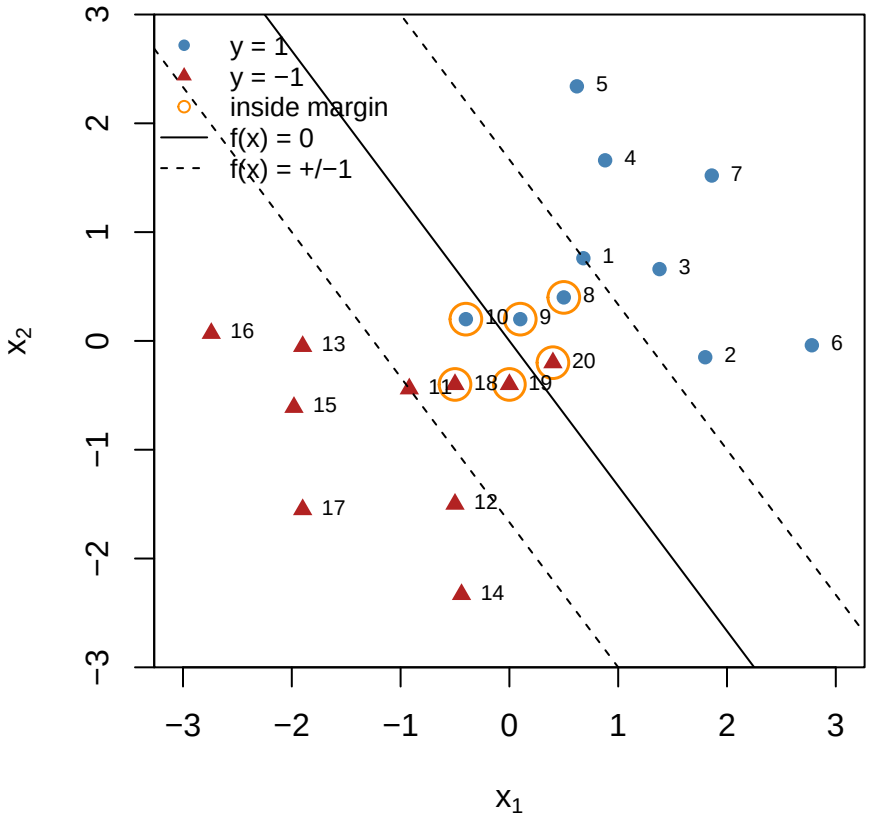
Question 1: Boundary and margin

```
d <- read.csv("data/margin-geometry-20.csv")
w <- c(0.8, 0.6)                                  # ||w|| = 1
d$f <- drop(as.matrix(d[, c("x1", "x2")]) %*% w)
d$distance <- abs(d$f) / sqrt(sum(w^2))           # distance to f(x) = 0
d$yf <- d$y * d$f                                 # functional margin
tol <- 1e-8
d$position <- ifelse(abs(d$yf - 1) < tol, "on",
                     ifelse(d$yf < 1, "inside", "beyond"))
d$misclassified <- d$yf < 0
knitr::kable(d[, c("observation", "x1", "x2", "y", "distance", "yf", "position", "misclassified")],
             digits = 2, row.names = FALSE)
```

observation	x1	x2	y	distance	yf	position	misclassified
1	0.68	0.76	1	1.00	1.00	on	FALSE
2	1.80	-0.15	1	1.35	1.35	beyond	FALSE
3	1.38	0.66	1	1.50	1.50	beyond	FALSE
4	0.88	1.66	1	1.70	1.70	beyond	FALSE
5	0.62	2.34	1	1.90	1.90	beyond	FALSE
6	2.78	-0.04	1	2.20	2.20	beyond	FALSE
7	1.86	1.52	1	2.40	2.40	beyond	FALSE
8	0.50	0.40	1	0.64	0.64	inside	FALSE
9	0.10	0.20	1	0.20	0.20	inside	FALSE
10	-0.40	0.20	1	0.20	-0.20	inside	TRUE
11	-0.92	-0.44	-1	1.00	1.00	on	FALSE
12	-0.50	-1.50	-1	1.30	1.30	beyond	FALSE
13	-1.90	-0.05	-1	1.55	1.55	beyond	FALSE
14	-0.44	-2.33	-1	1.75	1.75	beyond	FALSE
15	-1.98	-0.61	-1	1.95	1.95	beyond	FALSE
16	-2.74	0.07	-1	2.15	2.15	beyond	FALSE
17	-1.90	-1.55	-1	2.45	2.45	beyond	FALSE
18	-0.50	-0.40	-1	0.64	0.64	inside	FALSE
19	0.00	-0.40	-1	0.24	0.24	inside	FALSE
20	0.40	-0.20	-1	0.20	-0.20	inside	TRUE

Because $\|(0.8, 0.6)\|_2 = 1$, the distance to the boundary is $|f(x)|$. An observation is beyond its margin when $yf(x) > 1$, on it when $yf(x) = 1$, and inside it when $yf(x) < 1$.

```
cols <- ifelse(d$y == 1, "steelblue", "firebrick")
par(mar = c(4, 4, 1, 1), xaxs = "i", yaxs = "i")
plot(d$x1, d$x2, pch = ifelse(d$y == 1, 16, 17), col = cols, asp = 1,
     xlim = c(-3.2, 3.2), ylim = c(-3, 3), xlab = expression(x[1]), ylab = expression(x[2]))
for (k in c(-1, 0, 1))                              # 0.8 x1 + 0.6 x2 = k
  abline(a = k / 0.6, b = -0.8 / 0.6, lty = ifelse(k == 0, 1, 2))
ins <- d$position == "inside"
points(d$x1[ins], d$x2[ins], cex = 2.2, col = "darkorange", lwd = 1.5)
text(d$x1, d$x2, d$observation, pos = 4, cex = 0.7)
legend("topleft", c("y = 1", "y = -1", "inside margin", "f(x) = 0", "f(x) = +/-1"),
       pch = c(16, 17, 1, NA, NA), lty = c(NA, NA, NA, 1, 2),
       col = c("steelblue", "firebrick", "darkorange", "black", "black"), bty = "n", cex = 0.8)
```




```
table(factor(d$position, levels = c("inside", "on", "beyond")))
```

```
inside     on beyond
     6      2     12
```

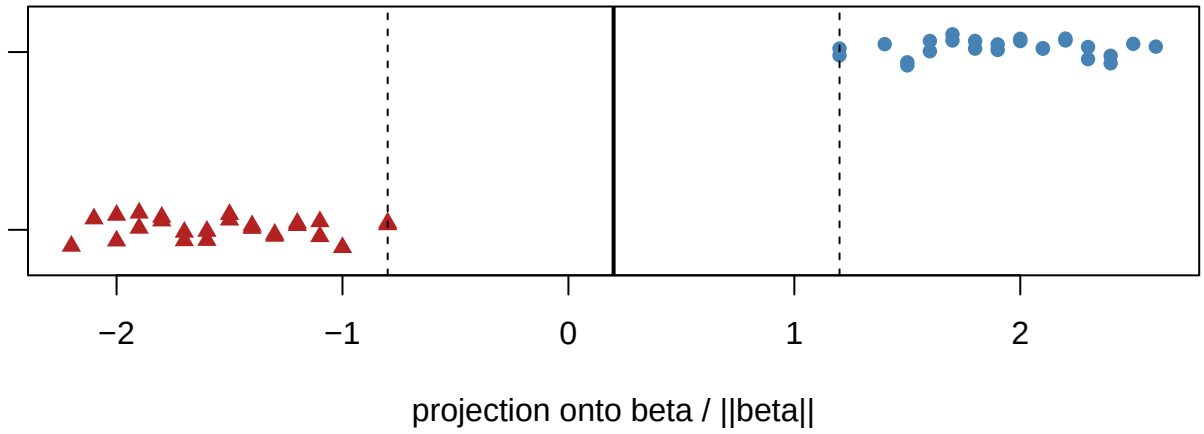
```
d$observation[d$misclassified]
```

```
[1] 10 20
```

Six observations (8, 9, 10, 18, 19, 20) are inside the margin, two (1 and 11) lie exactly on it, and the other twelve are beyond it. Observations 10 and 20 have $yf(x) < 0$, so they are on the wrong side of the boundary and are misclassified; the rest of the observations inside the margin are correctly classified margin violations.

Question 2: Margin in four dimensions

```
q <- read.csv("data/margin-geometry-4d.csv")
X <- as.matrix(q[, c("x1", "x2", "x3", "x4")])
beta0 <- -1; beta <- c(1, 2, -2, 4)
norm_b <- sqrt(sum(beta^2))
proj <- drop(X %*% beta) / norm_b                  # coordinate along beta / ||beta||
boundary <- -beta0 / norm_b                         # f(x) = 0 on this axis
signed_dist <- proj - boundary                      # = f(x) / ||beta||
par(mar = c(4, 1, 1, 1))
stripchart(list(`y = -1` = proj[q$y == -1], `y = 1` = proj[q$y == 1]), method = "jitter",
           pch = c(17, 16), col = c("firebrick", "steelblue"), xlab = "projection onto beta / ||beta||")
abline(v = boundary, lwd = 2)
abline(v = boundary + c(-1, 1) * min(q$y * signed_dist), lty = 2)
```




```
c(norm_beta = norm_b, boundary = boundary,
  M = min(q$y * signed_dist), all_correct = all(q$y * signed_dist > 0))
```

```
  norm_beta    boundary           M all_correct
        5.0         0.2         1.0         1.0
```

The solid line is the boundary at projection $-\beta_0/\|\beta\| = 0.2$. All observations are correctly classified, and the closest ones are a distance $M = 1$ from the boundary (dashed lines).

```
fm <- q$y * (beta0 + drop(X %*% beta))
c(min_functional_margin = min(fm))
```

```
min_functional_margin
                    5
```

```
c(beta0 = beta0, beta) / min(fm)
```

```
beta0
 -0.2   0.2   0.4  -0.4   0.8
```

The smallest functional margin under the given coefficients is $5 = \|\beta\|M$. Dividing by 5 gives the same boundary with $\min_i y_i f(\mathbf{x}_i) = 1$: $\beta = (0.2, 0.4, -0.4, 0.8)^\top$ and $\beta_0 = -0.2$. Then $\|\beta\| = 1$, consistent with $M = 1/\|\beta\| = 1$.

Question 3: Peer review of a Homework 7 skill

Assigned repository: mazy2005/stat432-fall2026 (Homework 7 report).

Running the skill

I copied the complete `SKILL.md` from the assigned Homework 7 report, unchanged, into a local folder (2,784 bytes; the report states 2,811 bytes, a difference I attribute to whitespace lost in the HTML rendering). I gave a fresh Claude Code agent this prompt:

> Follow the skill defined in `q3-run/SKILL.md` (read it first and treat it as your instructions) to analyze the SAheart data at `q3-run/saheart.csv`. Data information: the response is `chd` (0/1). The predictors are the other nine columns (sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, age). Code `famhist` numerically as Absent = 1, Present = 2. Work only inside `q3-run/`. Put the report there as report.pdf (or report.html if PDF rendering isn't possible), plus any code you write. Do not commit, push, or touch any other folder. Be efficient: aim to finish in about 15 minutes. When you finish, reply with: the files you produced, the seed, the candidates, the CV table, the selected model and its test error, and any problems you hit following the skill.

(Full paths were given in the actual prompt.) The run took about 81 seconds and about 39,000 tokens with six tool calls; no steer message was needed and the run was not stopped. Because the skill leaves the split to the invoking request, the agent chose a seed of 432, a stratified 75/25 split (346 training, 116 test), and 10 common stratified folds. It compared LDA, QDA, PCA(3)+LDA, PCA(3)+QDA, and a 50:50 LDA/QDA probability blend, selected ordinary LDA (CV error 0.266), and reported a test error of 0.276 (32/116). It built the PDF directly with matplotlib, and it said it rewrote a "main finding" sentence it had drafted before the results were in. The one-page report it produced:

SAheart: LDA vs. QDA classifiers for coronary heart disease (chd)

Data: n = 462 (chd=0: 302, chd=1: 160); no missing values; 9 numeric predictors (famhist coded Absent=1, Present=2). Full and per-class sample covariance matrices have full rank (9); no observations excluded.
Design (seed 432): stratified 75/25 split before any modelling -> training n = 346 (226/120), reserved test n = 116 (76/40). All candidates were scored on the same 10 stratified folds of the training set (StratifiedKFold, shuffle, random_state=432); scaling and PCA were refit inside each training fold. Criterion: mean held-out misclassification rate; exact ties go to the simpler model.

Figure 1. Cross-validation error by candidate (selected in red)

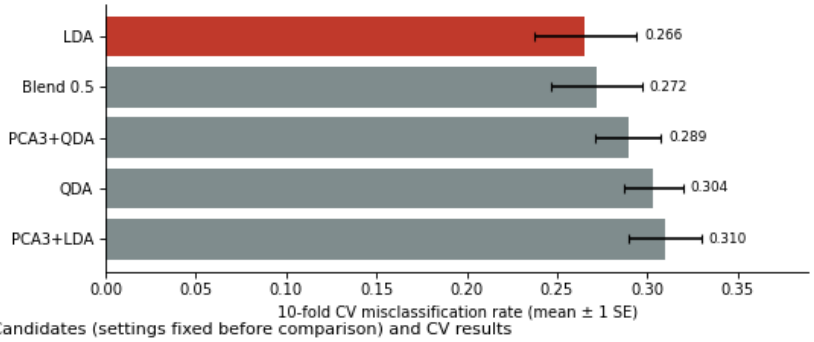


Table 1. Candidates (settings fixed before comparison) and CV results

Candidate	Settings	CV mean	Fold SD	SE
LDA	All 9 predictors, pooled covariance	0.266	0.090	0.028
QDA	All 9 predictors, class covariances	0.304	0.052	0.016
PCA3+LDA	Standardize, PCA k=3, then LDA	0.310	0.064	0.020
PCA3+QDA	Standardize, PCA k=3, then QDA	0.289	0.057	0.018
Blend 0.5	0.5*P(LDA) + 0.5*P(QDA), all 9	0.272	0.080	0.025

Selected model: LDA (All 9 predictors, pooled covariance). CV error 0.266 (SE 0.028). Refit on all 346 training cases and evaluated once on the 116 reserved test cases: test misclassification rate = 0.276 (32/116; approx. binomial SE 0.041).

Main finding: ordinary LDA had the lowest mean CV error, with the LDA/QDA probability blend close behind; ordinary QDA's extra class-specific covariance parameters did not help, and reducing to 3 principal components hurt LDA. Gaps between LDA, the blend and PCA3+QDA are within about one to two standard errors (folds are correlated, so SEs are approximate), so the ranking is not decisive.

Limitations: one random split and one fold assignment (results could shift with another seed); the test set is small (116 cases), so its error is a noisy estimate and was not used for any choice; LDA/QDA assume Gaussian class distributions, which skewed predictors (tobacco, alcohol) and the binary famhist violate; error rate ignores the unequal costs of missing CHD cases; SAheart is a retrospective sample of high-risk South African men, so results may not generalize. The smallest observed error does not establish general superiority of the selected method.
No covariance singularity or fit warnings occurred in any fold.

Reproducibility: analysis.py (this folder); full fold-level errors and diagnostics in results.json.

Peer review (written without AI assistance)

GitHub Issue: https://github.com/mazy2005/stat432-fall2026/issues/3

Overall, this is a solid analysis. The test set is held out before any model selection. All nine candidates use the same five folds, and scaling and PCA are fit inside each fold, so there is no leakage. The skill is general and only runs when explicitly invoked. I also liked that the report admits the ranking is not conclusive.

I ran the skill without your prompt. It finished in about 81 seconds, and I did not need to stop or steer it. It followed the procedure correctly, but it chose a different setup: seed 432, a 75/25 split, 10 folds, and 5 candidates. It also picked plain LDA instead of PCA-LDA with 6 components. My three comments come mostly from comparing the two runs.

1. The skill does not specify the study design. In your run, the split, fold count, PCA sizes, and blend weights

all came from your prompt. Without that prompt, my run made its own choices. It never tried 6 components, which is the setting that won for you. This matters because the skill is the deliverable. Two people running it on the same data got different final models. I suggest adding default settings to the skill, like a 30% test set, 5 folds, a set range of PCA components, and blend weights of 0.25, 0.5, and 0.75.

2. The selected model's lead is smaller than the uncertainty. PCA-LDA 6 scored 0.266, and the 75:25 blend scored 0.269. With 5 folds, the standard error for PCA-LDA 6 is about 0.017, which is bigger than that gap. Your text says the ranking is not conclusive, but the figure has no error bars, so it looks like a clear winner. I suggest adding standard error bars to the figure so readers can see how close the top models are.

3. The limitations leave out issues specific to this dataset. The report only mentions the close ranking and the single split. But famhist is binary and several predictors are skewed, so the normality assumption behind LDA and QDA does not hold. The data also come from a high-risk region, so the error rate may not carry over to other populations. I suggest adding a sentence or two on these points, since the skill asks for dataset-specific limitations.